CURRICULUM VITAE

PERSONAL INFORMATION	
Name	**AJEET KUMAR**
Date of birth	14/11/1989
Nationality	INDIAN
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E-mail	*ajeet.kumar.in@ieee.org ; ajeet.kumar@cnit.it ; akumar10@ec.iitr.ac.in*
Position	RESEARCHER LEVEL IV (PERMANENT APPOINTMENT)
Research Websites	Click 👉 Google Scholar Click 👉 Scopus Author ID: 57215268660 Click 👉 ResearcherID: AAQ-6015-2020 Click 👉 ORCID ID: 0000-0002-7758-8055

REFERENCES	Prof. Rajib Kumar Panigrahi, Professor, Department of Electronics and Communication Engineering, Indian Institute of Technology Roorkee, Roorkee 247667, India, ✉ rajib.panigrahi@ece.iitr.ac.in 📞 +91 7579467695	Dr. Anup Kumar Das Scientist/Engineer SG (Senior Grade) Space Applications Centre (SAC) Indian Space Research Organization (ISRO) Ahmedabad, Gujarat 380015, India ✉ anup@sac.isro.gov.in 📞 +91 9106276406
	Dr. Kumar Vijay Mishra, Senior Fellow / Principal Researcher, United States DEVCOM Army Research Laboratory (ARL) and The University of Maryland, College Park, USA, ✉ vizziee@gmail.com 📞 +1 (301) 675-5238	Dr. Raghu G. Raj, Head, Radar Imaging and Target ID Section, Code 5313, and Senior Research Scientist, U.S. Naval Research Laboratory, Washington D.C. 20375 ✉ raghu.g.raj.civ@us.navy.mil 📞 Office: 202-603-9808; Fax: 202-767-6276;

PROFILE

Brief description

Ajeet Kumar received his Ph.D. in RF and Microwave Engineering from the Department of Electronics and Communication Engineering, Indian Institute of Technology (IIT) Roorkee, India, in 2019. From 2014 to 2017, he served as a Research Fellow (Junior and Senior) on the Indian Space Research Organization's (ISRO) Radar Imaging Satellite (RISAT-1) mission. Subsequently, he worked as a Postdoctoral Research Associate on ISRO's Chandrayaan-2 space-borne mission for one year. In February 2021, Dr. Kumar joined the Radar and Surveillance Systems (RaSS) National Laboratory in Pisa as a CNIT researcher under contract. This position was converted to a permanent role in October 2023. Since 2021, he has been working on the Office of Naval Research Global (ONR Global) sponsored project of ATR by means of Polarimetric ISAR Images and multi-view 3D InISAR. Currently, he is also involved in another pioneer European Space Agency (ESA) "EO4SECURITY" project as a Responsabile Tecnico (Technical Lead) that is dedicated to monitoring environmental crimes and illegal activities. This project exploits the benefits of Inverse SAR to improve SAR imaging capabilities. Dr. Kumar's research interests include polarimetric SAR, hybrid-polarimetry SAR, interferometric SAR, and inverse SAR applications. He has been recently elevated to senior grade member of the International Union of Radio Science (URSI) for his vital contribution as a radio scientist. Also,

	he has been awarded two times the prestigious YSA (Young Scientist Award) given by the side of the International Union of Radio Science (URSI) in the year 2022 and 2024. He is currently an active member of the Synthetic Aperture Technical Working Group (SA-TWG) and serve as the Secretary of the IEEE P3355 Synthetic Aperture Radar Working Group (standardization activities related to synthetic aperture systems).
WORK EXPERIENCE	
(Oct. 2023 - Present)	Researcher IV (on Permanent Employment) at National RaSS (Radar and Surveillance System) Laboratory of National Inter-University Consortium for Telecommunications (CNIT, www.cnit.it/en), Pisa, Italy.
(Feb. 2021 – Sept. 2023)	Researcher IV (on temporary contract) at National RaSS (Radar and Surveillance System) Laboratory of National Inter-University Consortium for Telecommunications (CNIT, www.cnit.it/en), Pisa, Italy.
(Jan. 2020 - Jan. 2021)	Postdoctoral Research Associate in Chandrayaan-2 project sponsored by Space Applications Centre (SAC), ISRO, Ahmedabad, India
(Apr. 2017 - Oct. 2019)	Research Fellow in MHRD Institute Assistantship at IIT Roorkee.
(Mar. 2014 - Feb. 2017)	Research Fellow in ISRO Sponsored project of “Development of tools for Hybrid-Pol SAR data analysis and applications”
(Sept. 2013 - Feb. 2014)	Assistant Professor (on contract) at Buddha Institute of Technology, Bodh Gaya, India
EDUCATION	
(2014 - 2019)	Ph.D., Indian Institute of Technology Roorkee, India, RF and Microwave Engineering, Department of ECE (**1st division & 85% marks obtained**)
(2011 - 2013)	M. Tech., National Institute of Technology Patna, India Communication Systems, Department of ECE (**1st division & 82% Marks obtained**)
(2007 - 2011)	B. Tech., West Bengal University of Technology, India Department of Electronics and Communication Engineering (ECE) (**1st division & 8.24 CGPA obtained in 10.0 scale**)
LANGUAGE PROFICIENCY	Hindi: Native (Mother tongue) English: Fluent / Professional proficiency Italian: Basic (A1 level)
AWARDS, FELLOWSHIPS AND SCHOLARSHIPS	
	<u>Awards</u> ❖ Winner of URSI - Young Scientist Award 2024: I am recipient of the URSI-Young Scientist Award (YSA) 2024 that also covered the registration and accommodation cost to attend URSI-AT-RASC 2024 held in Gran Canaria, Spain. **(Visited Spain)** ❖ IEEE AESS Challenge Problem I - Radar-Based Heartbeat Monitoring (2025–2026, Funding Recipient) - Team selected as a

recipient of IEEE AESS funding under the “Call for Solutions: Radar-Based Heartbeat Monitoring” applied research challenge.

- IEEE AESS Radar Challenge 2025 - Won 3rd place as a member of the team that participated in the IEEE Aerospace & Electronic Systems Society Radar Challenge 2025.
- IEEE GRSS grant 2024: Won the partial travel grant from the IEEE GRSS society to attend IGARSS 2024 held in Athens, Greece. **(Visited Greece)**
- IEEE AESS YP Travel Grant: : Won the travel grant from the IEEE AESS society to attend International Radar Conference 2024 in Rennes, France. **(Visited France)**
- Winner of URSI - Young Scientist Award 2022: I am recipient of the URSI-Young Scientist Award (YSA) 2022 that also covered the registration and accommodation cost to attend URSI-RCRS 2022. **(Visited India)**
- IEEE GRSS grant: Won the travel grant from the IEEE GRSS society to attend IGARSS 2022 in Kuala Lumpur, Malaysia. **(Visited Malaysia)**
- IITR Alumni fund: Won the travel grant by Alumni Fund, IITR, Roorkee, India to attend POLINSAR 2019 conference held in Frascati, Italy. **(Visited Italy)**

Fellowships/Scholarship

- Post-Graduation MHRD fellowship: July 2011-June 2013
- Doctoral fellowship from ISRO: March 2014-March 2017
- Doctoral fellowship from MHRD: March 2017- July 2019
- As a part of PolInISAR based research project, awarded as a post-doctoral fellowship from CNIT, which is a National Inter-University Consortium for Telecommunications in Italy.

PARTICIPATION ON PROJECTS

Participation in the activities of a research group at national and international level

(Sept. 2023 - Present) @RaSS-CNIT, Italy	Responsabile Tecnico (Technical Lead) in European Space Agency (ESA) Funded Project. **Project Title:** “EO4Security: Innovative SAR processing methodology” **Funding Institution**: European Space Agency **Involved Country**: European Union. **Online Posting:** https://eo4society.esa.int/projects/eo4security-innovative-sar-processing-methodologies-for-security-applications-inverse-sar-processing-to-enhance-the-capabilities-to-characterize-targets-features-of-interest/
(April 2023 - Present) @RaSS-CNIT, Italy	Researcher IV in Ministry of Enterprises and Made in Italy Funded Project **Project Title:** “Tan-Tom: Diagnostic investigations on leathers using radiofrequency in EHF (30 – 300 GHz) bands” **Funding Institution**: Ministry of Enterprises and Made in Italy **Involved Country**: Italy.
(Feb. 2021 – Feb. 2024) @RaSS-CNIT, Italy	Researcher IV in Office of Naval Research (ONRG) sponsored project **Project Title:** “ATR by means of Polarimetric ISAR Images and multi-view 3D InISAR”

	Funding Institution: Office of Naval Research GLOBAL (ONRG) **Involved Country**: Italy, UK, USA. **Online Report Available**: https://apps.dtic.mil/sti/citations/trecms/AD1173222
(Jan. 2020 – Jan. 2021) *@IIT Roorkee, India*	Research Associate in ISRO sponsored project entitled: **Project Title:** “Development of Mathematical Models for Estimation of Lunar Regolith Roughness and Dielectric Constant using Chandrayaan-2 data.” **Funding Institution**: Indian Space Research organization **Involved Country**: India.
(Mar. 2014 – Feb. 2017) *@IIT Roorkee, India*	Research Fellow in ISRO sponsored project: **Project Title:** “Development of tools for hybrid-Pol SAR data analysis and applications” **Funding Institution**: Indian Space Research organization **Involved Country**: India

Academic Activities

Session Chair/Co-Chair in International Conferences

23rd Oct. 2024	**Conference Details:** International Conference Radar 2024 (IEEE), held in Rennes, France. **Session Name:** Emerging Trends in Machine Learning for Polarimetric and/or Interferometric SAR Imaging.
20th Dec.- 22th Dec. 2024	**Conference Details:** AIDCCSP 2024 - Artificial Intelligence, Device Computing, Communication, and Signal Processing (Springer), held in Hamirpur, India. **Session Name:** Session IV
17ht Aug. - 22nd Aug. 2025	**Conference Details:** URSI Asia-Pacific Radio Science Conference (URSI), held in Sydney, Australia. **Sessions Name:** **C01: Emerging Technologies for Radar & Communications** C05: Artificial Intelligence for Communications C09: Synthetic Aperture Radar (SAR) Signal Processing: Techniques and Applications C11: Short-Range Radars: Automotive, Biomedical, and Civilian Applications C12: Signal Processing for UAV-Based Systems C14: Array Signal Processing for Communications and Radar

Invited Planetary Lectures/Webinars

24th October 2025	Delivered an invited talk at University of Stirling on the topic: “*Compact Hybrid Polarimetry SAR for Earth, Planetary, and Military Applications”*
22nd Dec. 2024	Delivered an invited Plenary talk on the topic: “*Emerging Trends in Machine Learning for Polarimetric and/or Interferometric SAR Imaging”* Link: https://aidccsp24.nith.ac.in/program_Plenary_talks.html
8th May 2024	Delivered an invited IEEE Signal Processing Society (SPS) Webinar on the topic: *“Pol-InISAR Approach for 3D Imaging of Non-Cooperative Targets”*

	Link: https://signalprocessingsociety.org/blog/sps-webinar-pol-inisar-approach-3d-imaging-non-cooperative-targets
	Teaching assignments (during Ph.D.)
(Jan. 2019 – June 2019)	Radar Signal Processing
(Jan. 2018 – June 2018)	Radar Signal Processing
(July 2018 – Dec. 2018)	Microwave Engineering
(Jan. 2017 – June 2017)	Engineering Electromagnetics
	Participation in committees of scientific journals
(2023 - to - present)	Reviewer in the Journal IEEE Transactions on Geoscience and Remote Sensing
(2023 - to - present)	Reviewer in the Journal IEEE Antennas and Wireless Propagation Letters
(2023 - to - present)	Reviewer in the Journal IEEE Transactions on Computational Imaging
(2020 - to - present)	Reviewer in the Journal IEEE Geoscience and Remote Sensing Letters
(2023 - to - present)	Reviewer in the Journal IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing
(2023 - to - present)	Reviewer in the IEEE Signal Processing Letters
(2022 - to - present)	Reviewer in the Journal Remote Sensing (MDPI)
(2018 - to - present)	Reviewer in the Journal IET Radar, Sonar, and Navigation
(2021 - to - present)	Reviewer in the Journal IET Electronics Letters
(2021 - to - present)	Reviewer in the Journal Advances in Space Research (Elsevier)
(2022 - to - present)	Reviewer in the Journal Environmental Technology (Taylor & Francis)
	Organizing Conferences, Workshops and Seminars
Dec. 15-17, 2017	Technical Coordinator at 14th IEEE India Council International Conference INDICON 2017
Dec. 03-04, 2016	Technical Coordinator at 11th International Conference on Industrial and Information Systems ICIIS 2016
Feb. 13-15, 2015	Technical Coordinator at National Conference on Recent Advances in Electronics & Computer Engineering RAECE 2015
Nov. 17-19, 2017	Technical Coordinator at National Workshop on Vacuum Electron Devices & its Applications (VEDA-2017)
June 08-09, 2023	Technical Program Committee member at International Conference on Computer, Electronics and Electrical Engineering and Their Applications (IC2E3-2023)
July 22-23, 2024	Technical Program Committee member at the International IEEE Conference "SPACE 2024"
PUBLICATIONS	

SCI Journals

1.		**Ajeet Kumar**, I. M. Kochar, D. K. Pandey, A. Das, D. Putrevu, R. Kumar, R. K. Panigrahi, "Dielectric Constant Estimation of Lunar Surface Using Mini-RF and Chandrayaan-2 SAR Data," in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1-8, Art no. 4600608, 2022. DOI: 10.1109/TGRS.2021.3103383. (Impact Factor: 8.2)
2.		I. Kochar, H. Maurya, **Ajeet Kumar**, S. S. Bhiravarasu, A. Das, D. Putrevu, D. K. Pandey, R. K. Panigrahi, "Retrieval of Lunar Surface Dielectric Constant using Chandrayaan-2 Full-Polarimetric SAR Data," in *IEEE Transactions on Geoscience and Remote Sensing,* vol. 60, pp. 1-12, Art no. 4602212, 2022. DOI: 10.1109/TGRS.2022.3201050. (Impact Factor: 8.2)
3.		**Ajeet Kumar**, E. Giusti, F. Mancuso, S. Ghio, A. Lupidi, and M. Martorella "Three-Dimensional Polarimetric InISAR Imaging of Non-Cooperative Targets," in *IEEE Transactions on Computational Imaging,* vol. 9, pp. 210-223, 2023. DOI: 10.1109/TCI.2023.3248942 (Impact Factor: 5.4)
4.		**Ajeet Kumar**, V. Mishra, R. K. Panigrahi, and M. Martorella "Application of Hybrid-pol SAR in Oil-spill Detection," in *IEEE Geoscience and Remote Sensing Letters,* vol. 20, pp. 1-5, 2023, Art no. 4004505. DOI: 10.1109/LGRS.2023.3258224 (Impact Factor: 4.8)
5.		**Ajeet Kumar**, A. Das, and R. K. Panigrahi, "Hybrid-Pol Based Three-Component Scattering Model for Analysis of RISAT Data," in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 10, no. 12, pp. 5155-5162, Dec. 2017. DOI: 10.1109/JSTARS.2017.2768378 (Impact Factor: 5.5)
6.		**Ajeet Kumar**, A. Das, and R. K. Panigrahi, "Hybrid-pol Decomposition Methods: A Comparative Evaluation and a New Entropy-based Approach*,"* in *IETE Technical Review*, 37:3, 296-308, 2020. DOI: 10.1080/02564602.2019.1613937.
7.		**Ajeet Kumar** and R. K. Panigrahi, "Entropy based reconstruction technique for analysis of hybrid-polarimetric SAR data," in *IET Radar, Sonar & Navigation*, vol. 13, no. 4, pp. 620-626, Apr. 2019. DOI: 10.1049/iet-rsn.2018.5338
8.		**Ajeet Kumar**, R. K. Panigrahi, and A. Das, "Three-component decomposition technique for hybrid-pol SAR data," in *IET Radar, Sonar & Navigation*, vol. 10, no. 9, pp. 1569-1574, Dec. 2016. DOI: 10.1049/iet-rsn.2015.0298
9.		**Ajeet Kumar** and R. K. Panigrahi, "Oil-spill detection using hybrid-pol data through reconstruction technique," in *Electronics Letters*, vol. 55, no. 9, pp. 558-561, May 2019. DOI: 10.1049/el.2018.6914
10.		**Ajeet Kumar** and R. K. Panigrahi, "Classification of hybrid-pol data using novel cross-polarisation estimation approach," in *Electronics Letters*, vol. 54, no. 3, pp. 161-163, Feb. 2018. DOI: 10.1049/el.2017.3855
11.		S. Awasthi, D. Varade, P. K. Thakur, **Ajeet Kumar**, H. Singh, K. Jain, Snehmani, "Development of a novel approach for snow wetness estimation using hybrid polarimetric RISAT-1 SAR datasets in North-Western Himalayan region," in *Journal of Hydrology*, Vol. 612, 128252, 2022. DOI: 10.1016/j.jhydrol.2022.128252. (Impact Factor: 6.7)
12.		S. Awasthi, K. Jain, V. Mishra, **Ajeet Kumar**, "An approach for multi-dimensional land subsidence velocity estimation using time-series Sentinel-1 SAR datasets by applying persistent scatterer interferometry technique," in *Geocarto International*, 37:9, 2647-2678, Oct. 2020. DOI: 10.1080/10106049.2020.1831624. (Impact Factor: 3.5)

13.	S. Awasthi, P. K. Thakur, S. Kumar, **Ajeet Kumar**, K. Jain and **Snehmani**, “Snow Density Retrieval Using Hybrid Polarimetric RISAT-1 Datasets,” in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 13, pp. 3058-3065, 2020. DOI: 10.1109/JSTARS.2020.2991156. (Impact Factor: 5.5)
14.	K. Inderkumar, **Ajeet Kumar** and R. K. Panigrahi, “Three-Component Decomposition of Hybrid-PolSAR Data with Volume Scattering Models,” in *IEEE Access*, vol. 9, pp. 117415-117423, 2021. DOI: 10.1109/ACCESS.2021.3106151. (Impact Factor: 3.9)
15.	S. Awasthi, S. Kumar, P. K. Thakur, K. Jain, **Ajeet Kumar** & Snehmani, “Snow depth retrieval in North-Western Himalayan region using pursuit-monostatic TanDEM-X datasets applying polarimetric synthetic aperture radar interferometry-based inversion Modelling,” in *International Journal of Remote Sensing*, 42:8, 2872-2897, DOI: 10.1080/01431161.2020.1862439. (Impact Factor: 3.4)
16.	D. Trivedi, G. S. Phartiyal, **Ajeet Kumar**, D. Singh, “Synthetic Imaging Radar Data Generation in Various Clutter Environments Using Novel UWB Log-Periodic Antenna,” in *Sensors ,* 2024; 24(24):7903. DOI: 10.3390/s24247903. (Impact Factor: 3.5)
17.	G. Meucci, E. Giusti, **Ajeet Kumar**, F. Mancuso, S. Ghio and M. Martorella, “Transformer-based Automatic Target Recognition for 3D-InISAR,” in *IEEE Transactions on Radar Systems,* vol. 3, pp. 180-192, 2025, DOI: 10.1109/TRS.2025.3527281.
18.	A. Bhatt, **Ajeet Kumar**, and T. Goel, "Modified Hybrid-Pol Three-Component Scattering Decomposition Model," in *IEEE Geoscience and Remote Sensing Letters*, vol. 22, pp. 1-5, 2025, Art no. 4005005, DOI: 10.1109/LGRS.2025.3538217. (2025 Impact Factor: 4)
19.	**Ajeet Kumar**, E. Giusti, M. Martorella “Hybrid polarimetry inverse synthetic aperture radar,” *IET Radar, Sonar, and Navigation,* e70004 (2025). DOI: doi.org/10.1049/rsn2.70004
20.	B. Ahuja, L. Gentile, **Ajeet Kumar**, M. Martorella “Role of Radio Telescopes in Space Debris Monitoring: Current Insights and Future Directions,” *Sensors,* **2025**, 25, 2900. (2025 Impact Factor: 3.5) DOI: doi.org/10.3390/s25092900
21.	N. Kumar, VS. Rathore, **Ajeet Kumar**, A. P. Krishna, R. Biswas, A. K. Das, "Morphological characterization of impact craters in the south polar region of Moon using Chandrayaan-2 Dual Frequency Synthetic Aperture Radar data," in *Advances in Space Research*, 2025. DOI: https://doi.org/10.1016/j.asr.2025.10.060. (2025 Impact Factor: 2.8)

*** Impact Factor reflects past values and may not accurately reflect current changes, as they vary on a yearly basis**

SCI Journals Summary:

IEEE Transactions: 4
IET Journals: 5
IEEE Letters: 2
Other SCI Journals: 7
IEEE Journals (other than Transactions and Letters): 3

Total no. of SCI Journals: 21
(First Author SCI Journals: 10)

Book Chapters

1.	E. Giusti, C. Y. Pui, **Ajeet Kumar**, S. Ghio, B. Ng, L. Rosenberg, M. Martorella, and T. Cao “Polarimetric Three-Dimensional Inverse Synthetic Aperture Radar,” in **Multidimensional Radar Imaging, Volume 2**, *The Institution of Engineering and Technology (IET)*, pp. 159–184, Chapter 7, 2024.

	DOI: **10.1049/SBRA562E_ch7**
	Published Technical Report
1.	M. Martorella, E. Giusti, **Ajeet Kumar**, F. Mancuso, G. Meucci, A. Lupidi, and S. Ghio "ATR by means of polarimetric ISAR images and multiview 3D InISAR," in *Defense Technical Information Center (DTIC)*, pp. 159–184, June 2022 [Online at: https://apps.dtic.mil/sti/pdfs/AD1173222.pdf].
	Selected International Conferences
1.	**Ajeet Kumar**, E. Giusti, and M. Martorella, "A Novel Back-Projection Based ISAR Approach for Non-Cooperative Moving Targets in SAR Images," in *2025 IEEE Radar Conference (RadarConf25)*, Oct. 2025, pp. 1451-1455, Krakow, Poland.
2.	E. Giusti, **Ajeet Kumar**, and M. Martorella, "Inverse SAR Pulse-to-Pulse Phase Correction for Improving SAR Imaging of Non-Cooperative Targets," in *2025 IEEE Radar Conference (RadarConf25)*, Oct. 2025, pp. 1005-1010, Krakow, Poland.
3.	**Ajeet Kumar** and R. K. Panigrahi, "Classification of hybrid-pol data based on Euclidean distance between Stokes vectors," in *2015 IEEE Radar Conference (RadarConf15)*, Oct. 2015, pp. 422-425, Johannesburg, South Africa.
4.	**Ajeet Kumar**, A. Das, and R. K. Panigrahi, "A comparative analysis of hybrid-pol algorithms using RISAT data," in *2016 IEEE Asia-Pacific Microwave Conference (APMC)*, Dec. 2017, pp. 1- New Delhi, India. doi: 10.1109/APMC.2016.7931415
5.	**Ajeet Kumar**, R. Sharma, A. Das, and R. K. Panigrahi, "RISAT data analysis for land-cover classification," in *11th International Radar Symposium India 2017 (IRSI-17)*, Dec. 2017, Bangalore, India.
6.	**Ajeet Kumar**, A. Das, and R. K. Panigrahi, "Hybrid-pol data analysis for land cover classification," in *Proceeding POLinSAR 2019*, Jan. 2019, Rome, Italy.
7.	R. Sharma, **Ajeet Kumar**, and R. K. Panigrahi, "Investigating the implementation issues of speckle filters for single look hybrid-polSAR data," in *Proceeding POLinSAR 2019*, Jan. 2019, Rome, Italy.
8.	**Ajeet Kumar**, A. Das, and R. K. Panigrahi, "Analysis using dielectric variation-based hybrid-pol three-component model-based method," in *IEEE XXXIIIrd General Assembly and Scientific Symposium of the International Union of Radio Science*, 2020, pp. 1-4, Rome, Italy. doi: 10.23919/URSIGASS49373.2020.9232030
9.	**Ajeet Kumar**, E. Giusti, F. Mancuso and M. Martorella, "Polarimetric Interferometric ISAR Based 3-D Imaging of Non-Cooperative Target," in *IGARSS 2022 - 2022 IEEE International Geoscience and Remote Sensing Symposium*, 2022, pp. 385-388, Kuala Lumpur, Malaysia. doi: 10.1109/IGARSS46834.2022.9883669
10.	E. Giusti, **Ajeet Kumar**, F. Mancuso, S. Ghio and M. Martorella, "Fully polarimetric multi-aspect 3D InISAR," *2022 23rd International Radar Symposium (IRS),* Gdansk, Poland, 2022, pp. 184-189, doi: 10.23919/IRS54158.2022.9905018.
11.	F. Mancuso, E. Giusti, **Ajeet Kumar**, S. Ghio and M. Martorella, "Comparative assessment of polarimetric features estimation in fully polarimetric 3D-ISAR imaging system," *International Conference on Radar Systems (RADAR 2022)*, Hybrid Conference, Edinburgh, UK, 2022, pp. 353-358, doi: 10.1049/icp.2022.2343.
12.	**Ajeet Kumar**, V. Mishra, R. K. Panigrahi, M. Martorella, "Hybrid-polarimetry Synthetic Aperture Radar for Oil-Spill Detection," *2022 URSI Regional*

		Conference on Radio Science (USRI-RCRS), Indore, India, 2022, pp. 1-4, doi: 10.23919/URSI-RCRS56822.2022.10118472.
13.		G. Meucci, F. Mancuso, E. Giusti, **Ajeet Kumar**, S. Ghio and M. Martorella, "Point Cloud Transformer (PCT) for 3D-InISAR Automatic Target Recognition," *2023 IEEE Radar Conference (RadarConf23),* San Antonio, TX, USA, 2023, pp. 1-6, doi: 10.1109/RadarConf2351548.2023.10149787..
14.		A. Bhatt, S. Pathak, M. K. Aghwariya, P. K. Pal, **Ajeet Kumar** and T. Goel, "Model Based Decomposition of Alakananda River Basin using SAR Data," *2023 First International Conference on Microwave, Antenna and Communication (MAC),* Prayagraj, India, 2023, pp. 1-6.
15.		**A. Kumar**, E. Giusti and M. Martorella, "Hybrid Polarimetry Inverse SAR," *2023 IEEE International Radar Conference (RADAR),* Sydney, Australia, 2023, pp. 1-6, doi: 10.1109/RADAR54928.2023.10371162
16.		**A. Kumar**, R. K. Panigrahi and M. Martorella, "Accurate Estimation of $\langle \lvert S_{HV} \rvert^2 \rangle$ in Hybrid-polarimetry SAR: A closed-form solution," *IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium,* Athens, Greece, 2024, pp. 10786-10789, doi: 10.1109/IGARSS53475.2024.10642857
17.		**A. Kumar,** R. K. Panigrahi and M. Martorella**,** "Closed-Form Solution for Accurate Estimation of $\langle \lvert S_{HV} \rvert^2 \rangle$ In Hybrid-Polarimetry SAR For Oil Spill Monitoring**,"** *2024 4th URSI Atlantic Radio Science Meeting (AT-RASC),* Meloneras, Spain, 2024, pp. 1-4, doi: 10.46620/URSIATRASC24/SHIA5543.
18.		A. Oveis**, A. Kumar,** E. Giusti, and A. Cantelli**,** "Adversarial Perturbations in Pol-ISAR Image Classification: A Study on Explanation Consistency and Ensemble Learning**,"** *2024 IEEE International Radar Conference (RADAR),* Rennes, France, 2024.
19.		G. S. Phartiyal**, A. Kumar,** " Interpreting Oil-Spill Mapping DNN: A Case Study with Hybrid-pol SAR data**,"** *2024 IEEE International Radar Conference (RADAR),* Rennes, France, 2024.
20.		N. Kumar, V. S. Rathore, A. P. Krishna, **A. Kumar**, R. Biswas, A. K. Das**,** " Backscatter Coefficients and Circular Polarization Ratio based Study of Impact Craters in the Permanently Shadowed Region of the Lunar South Pole using Chandrayaan-2 DFSAR data**,"** *2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS),* Goa, India, 2024, pp. 1-4. doi: 10.1109/InGARSS61818.2024.10984313
21.		**A. Kumar**, A. H. Oveis and M. Martorella**,** " Synthetic Aperture Radar for Oil Spill Detection and Characterization: Special Focus on Arctic Routes**,"** *2025 22nd European Radar Conference (EuRAD),* Utrecht, Netherlands, 2025, pp. 270-273 doi: 10.23919/EuRAD65285.2025.11234181
22.		A. Bhatt, **Ajeet Kumar**, and T. Goel, "Monitoring Land Deformation Using DInSAR: Insights From Sentinel-1 Data," in *URSI Radio Science Letters,* vol. 7, pp. 1-5, 2025, doi: 10.46620/25-0030. (Conference Paper Promoted to Journal)
23.		**A. Kumar** and M. Martorella, "Future Oil Spill Risks in Arctic Shipping Routes and Satellite-SAR-Based Solution for Oil Spill Detection and Characterization," in *URSI Asia-Pacific Radio Science Meeting (AP-RASC) 2025,* Sydney, Australia, 2025, pp. 1-4. doi: 10.46620/URSIAPRASC25/OZUJ9186.

I hereby express my consent to process and use my data provided in this curriculum vitae.

Pisa, 30/03/2026 Ajeet Kumar

Ajeet Kumar

Place, date	Signature